MODULE 6

Rotation, Pose and Frames

6.1 Pose = where you are + which way you face

SIMPLE APPROACH

Stand in the middle of a room. "Where you are" is your spot on the floor (and how tall you are). "Which way you face" is whether you look at the board or the window, whether you tilt your head, whether you lean. A LiDAR on a car needs to know both of these things, all the time, because the car bumps, turns and leans. Together they are called its **pose**.

ENGINEER'S VIEW

Pose has six degrees of freedom: position $\mathbf{p} = (X, Y, Z)$ and orientation, usually written as three angles — **roll** ϕ, **pitch** θ, **yaw** ψ — or as a 3×3 rotation matrix $\mathbf{R}$, or a unit quaternion. In LIO-SAM's notation the pose is a rigid-body transformation $\mathbf{T} = [\mathbf{R} \mid \mathbf{p}] \in SE(3)$, where $SE(3)$ ("special Euclidean group") is simply the set of all rotations-plus-translations, and $\mathbf{R} \in SO(3)$ ("special orthogonal group") the set of all proper rotation matrices.

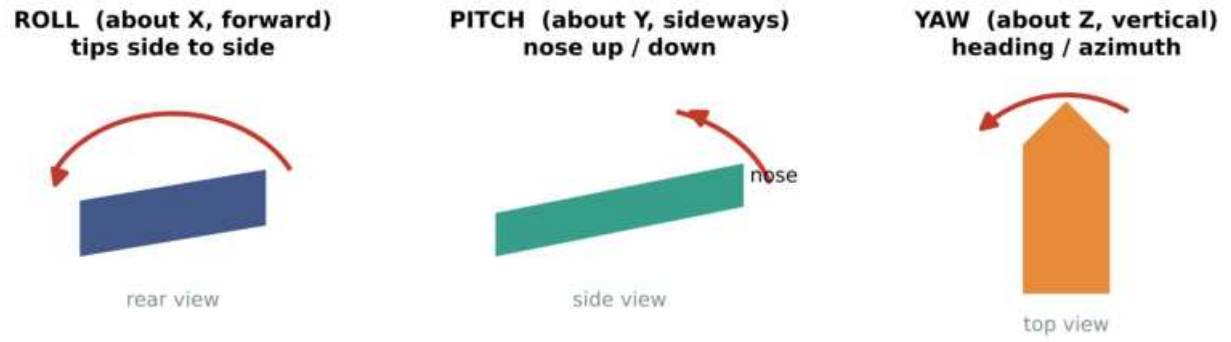


Figure 6.1: Roll, pitch and yaw on a mapping vehicle.

Angle	About axis	Everyday example	Effect of a residual error on the map
Roll ϕ	x (forward)	banked curve	tilts points left/right of the drive line; heights wrong on one side
Pitch θ	y (sideways)	climbing a grade	far points too high/low ahead of and behind the vehicle
Yaw ψ	z (vertical)	turning a corner	rotates the whole scan; lateral error grows with range

6.2 The right-hand rule: naming X, Y, Z and roll, pitch, yaw

SIMPLE APPROACH

Hold up your **right** hand. Point your **first (index) finger** straight ahead — that is **X**. Point your **middle finger** to the left — that is **Y**. Stick your **thumb** straight up — that is **Z**. **The thumb is always Z.** Another way to say it: curl your fingers from X toward Y, and your thumb points to Z. Now make a thumbs-up: the way your curled fingers wrap around your thumb is the "plus" direction for turning around that thumb. Point your thumb along X and your fingers show plus **roll**; along Y, plus **pitch**; along Z (straight up), plus **yaw**.

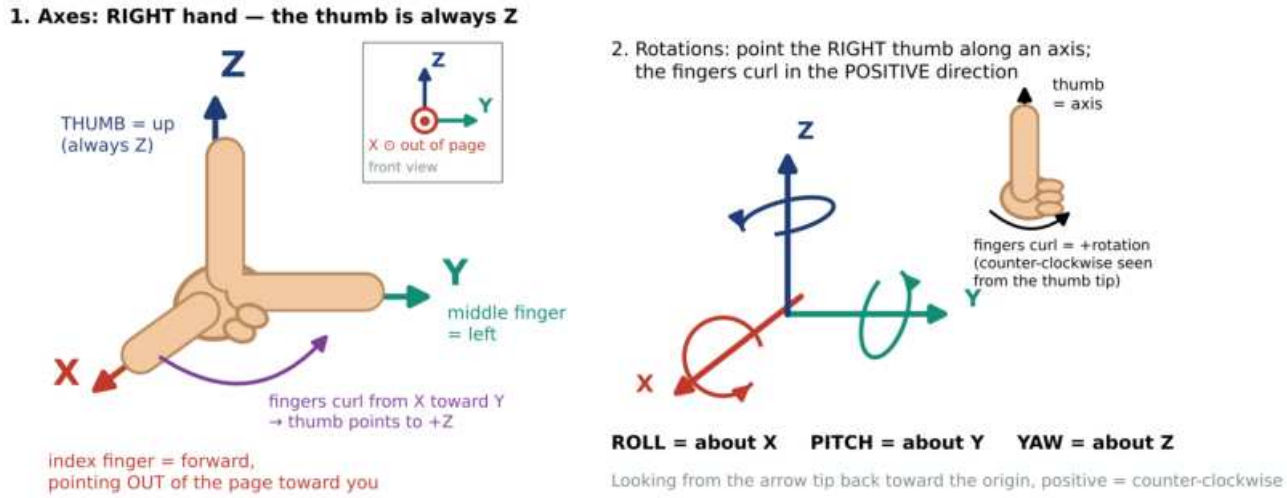


Figure 6.2: The right-hand rule. Left: index finger = X, middle finger = Y, and the thumb is always Z; equivalently, the fingers curl from X toward Y and the thumb points to Z. Right: with the thumb along an axis, the curled fingers show the positive rotation about it — roll about X, pitch about Y, yaw about Z.

ENGINEER'S VIEW

Rule 1 — the axes form a right-handed set. With the right hand: index finger $= +X$, middle finger $= +Y$, **thumb** $= +Z$ — the thumb is always Z. Equivalently, curl the fingers from $+X$ toward $+Y$ and the thumb points along $+Z$; mathematically $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$ (Module 2). (Some textbooks label the fingers in a different order, e.g. thumb $= X$; any cyclic relabeling gives the same right-handed set, but this course keeps the thumb on Z throughout so that axis naming and the rotation rule below use the hand the same way.) Every coordinate frame in this course — sensor, body, IMU, world (ENU) and map projection with height — is right-handed. A left-handed frame (one axis flipped) is a mirror image, and no rotation can turn one into the other; mixing them is a classic cause of a "mirrored" point cloud.

Rule 2 — positive rotations follow the curl. Point the right thumb along the positive axis; the fingers curl in the direction of a *positive* angle. Seen from the arrow tip looking back toward the origin, positive is **counter-clockwise**.

- **Roll** ϕ = rotation about $+X$
- **Pitch** θ = rotation about $+Y$
- **Yaw** ψ = rotation about $+Z$

Because the *direction* of each axis decides the sign, the same physical motion can be $+$ in one convention and $-$ in another. The two conventions you will meet most:

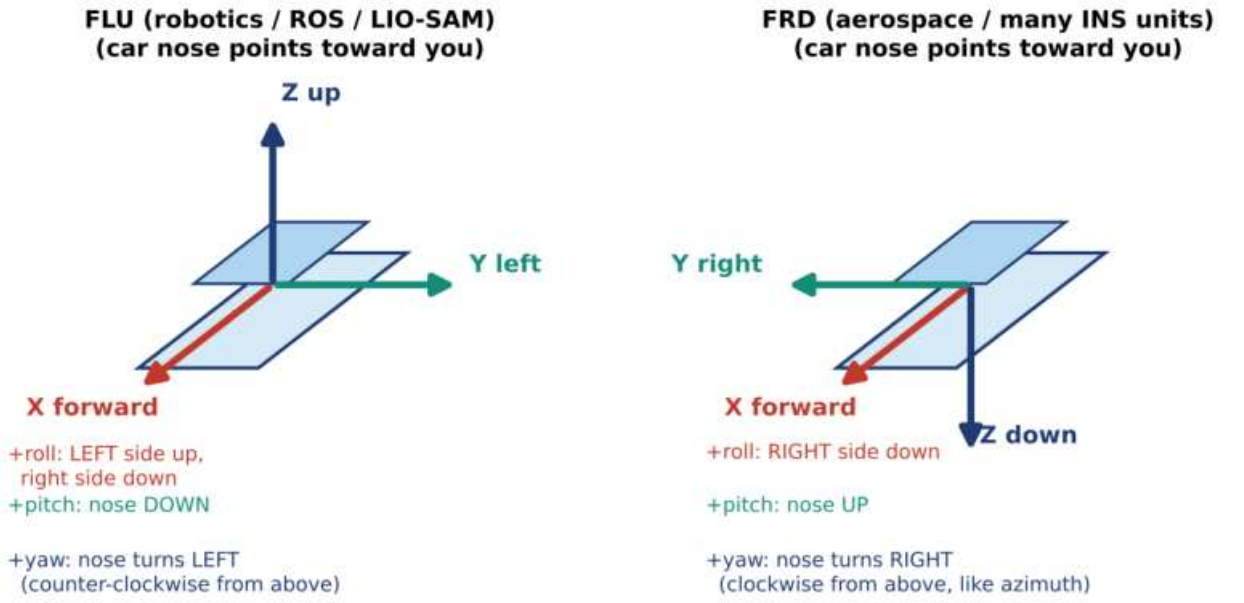


Figure 6.3: The same vehicle in the two common right-handed body frames.

	FLU: X forward, Y left, Z up	FRD: X forward, Y right, Z down
Used by	ROS (REP-103), LIO-SAM, most LiDAR drivers, this course	aircraft, many INS/IMU products (NED world frame)
+ roll (about X)	left side *up*, right side down	right side *down*
+ pitch (about Y)	nose *down*	nose *up*
+ yaw (about Z)	nose turns *left* (counter-clockwise from above)	nose turns *right* (clockwise from above, like azimuth)
World frame	ENU: East, North, Up	NED: North, East, Down

WORKED EXAMPLE — checking a frame with your hand (and with the cross product)

Given: a datasheet says a sensor's axes are "X forward, Y right, Z up."

Hand test

right index finger forward (X), middle finger right (Y) — the thumb (Z) now points *down*, not up.

Cross-product test

forward $\times$ right $= (1,0,0) \times (0,-1,0) = (0,0,-1)$ = down in a frame where $(0,1,0)$ is left.

Result: "X forward, Y right, Z up" is **left-handed**. Either the datasheet means Z down (FRD) or one axis is mislabeled. Resolve it with the manufacturer before computing a single lever arm or boresight; a sign error here mirrors or tilts the entire cloud.

SURVEYOR'S FIELD TIP

When setting up a rig, stand behind the vehicle, hold up your right hand in the way the manual defines the axes, and *physically* roll, pitch and yaw the IMU a few degrees while watching the live output. Each angle should increase in the direction your fingers curl. Write the verified convention in the calibration record (Appendix E).

6.3 Rotating a point in 2-D

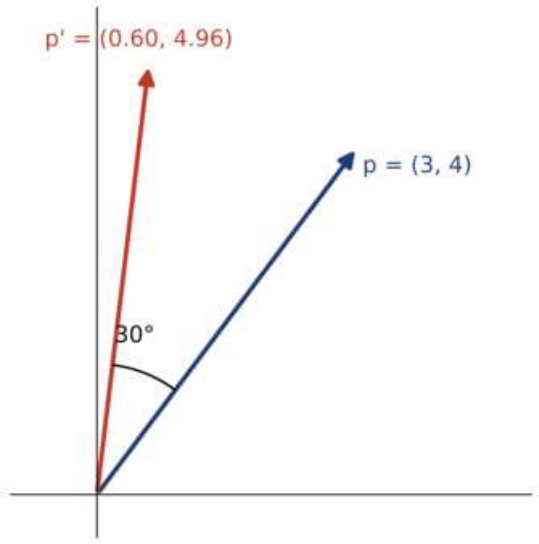


Figure 6.4: Rotating $\mathbf{p} = (3, 4)$ by 30° counter-clockwise.

KEY EQUATION — 2-D rotation

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \underbrace{\begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix}}_{\mathbf{R}(\psi)} \begin{pmatrix} x \\ y \end{pmatrix}$$

What every symbol means

Symbol	Meaning	Units
(x, y)	original point	m
(x', y')	point after rotation	m
ψ	rotation angle, positive counter-clockwise ("psi")	rad
$\mathbf{R}(\psi)$	the rotation matrix; its columns are where the x- and y-axes end up	—

WORKED EXAMPLE — rotate (3, 4) by 30°

Step 1

$\cos 30^\circ = 0.86603$, $\sin 30^\circ = 0.50000$

Step 2

$x' = 0.86603 \times 3 - 0.5 \times 4 = 2.59808 - 2.00000 = 0.598$

Step 3

$y' = 0.5 \times 3 + 0.86603 \times 4 = 1.50000 + 3.46410 = 4.964$

Step 4 — check

length before $= \sqrt{9+16} = 5$; after $= \sqrt{0.598^2 + 4.964^2} = \sqrt{0.358 + 24.642} = 5.000$ ✓

Result: $(0.598,\ 4.964)$. A rotation never changes lengths — a great error check.

6.4 Rotations in 3-D

KEY EQUATION — The three elementary rotations

$$\mathbf{R}_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{pmatrix},\ \mathbf{R}_y(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix},\ \mathbf{R}_z(\psi) = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{R} = \mathbf{R}_z(\psi)\,\mathbf{R}_y(\theta)\,\mathbf{R}_x(\phi) \qquad (\text{“ZYX” or yaw–pitch–roll order})$$

What every symbol means

Symbol	Meaning	Units
ϕ, θ, ψ	roll, pitch, yaw (“phi, theta, psi”)	rad
$\mathbf{R}_x$ etc.	rotation about one axis only; the axis’s own row and column hold 1 and 0s	—
$\mathbf{R}$	the full body-to-world rotation; applied right-to-left: roll first, then pitch, then yaw	—

ENGINEER’S VIEW

Order matters. Rotating 90° about x then 90° about z ends somewhere different than the reverse order — try it with a book. Every software package documents its order and sign convention; a large share of real-world georeferencing bugs are convention mismatches. Two other important facts:

- $\mathbf{R}^\mathsf{T}\mathbf{R} = \mathbf{I}$ and $\det \mathbf{R} = +1$. So the inverse rotation (world-to-body) is just $\mathbf{R}^\mathsf{T}$. LIO-SAM writes $\mathbf{R}_t = \mathbf{R}_t^{\mathsf{WB}} = \mathbf{R}_t^{\mathsf{BW}\mathsf{T}}$ for exactly this reason.
- In the FLU frame used here (x forward, y left, z up), a *positive* rotation about y tips the nose *down*. Many INS products use FRD (x forward, y right, z down) where positive pitch is nose up. Always confirm.

SURVEYOR’S FIELD TIP

Yaw is not azimuth. Surveyors measure azimuth clockwise from grid north. ROS/ENU systems measure yaw counter-clockwise from east. Convert with $\psi = 90^\circ - \alpha_{\text{az}}$. A vehicle heading due north has $\alpha_{\text{az}} = 0^\circ$ but $\psi = 90^\circ$. Grid north also differs from true north by the *convergence angle* of your map projection — another place a heading can quietly go wrong.

6.5 Homogeneous transforms: rotation and shift in one matrix

KEY EQUATION — Rigid-body transformation

$$\mathbf{r}_W = \mathbf{R}\,\mathbf{r}_B + \mathbf{p} \quad \Longleftrightarrow \quad \begin{pmatrix} \mathbf{r}_W \\ 1 \end{pmatrix} = \underbrace{\begin{pmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{pmatrix}}_{\mathbf{T}} \begin{pmatrix} \mathbf{r}_B \\ 1 \end{pmatrix}$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{r}_B$	a point expressed in the body (vehicle/IMU) frame B	m
$\mathbf{r}_W$	the same point expressed in the world frame W	m
$\mathbf{R}$	body-to-world rotation	—
$\mathbf{p}$	position of the body origin in W	m
$\mathbf{T}$	4×4 transform; chaining frames = multiplying $\mathbf{T}$'s	—

The relative motion between two poses is $\Delta\mathbf{T}_{i,j} = \mathbf{T}_i^{-1}\mathbf{T}_j$. That is LIO-SAM's equation (12), which defines the *lidar odometry factor* (Module 12). (The paper prints it as $\mathbf{T}_i^{\mathsf{T}}\mathbf{T}_{i+1}$; for the full 4×4 matrix the correct operation is the inverse, which equals the transpose only for the rotation block.)

6.6 Rotation from a turning rate: the exponential map

Gyroscopes give angular *rates*. Over a short time Δt the body turns through a rotation vector $\boldsymbol{\omega}\Delta t$ (direction = axis, length = angle). LIO-SAM's equation (6) converts that small turn into a rotation matrix with the exponential map:

KEY EQUATION — Rotation update (Rodrigues' formula)

$$\mathbf{R}_{t+\Delta t} = \mathbf{R}_t \exp\big([\boldsymbol{\omega}\Delta t]_\times\big), \qquad \exp([\mathbf{a}]_\times) = \mathbf{I} + \frac{\sin\alpha}{\alpha}[\mathbf{a}]_\times + \frac{1-\cos\alpha}{\alpha^2}[\mathbf{a}]_\times^2$$

What every symbol means

Symbol	Meaning	Units
$\boldsymbol{\omega}$	angular-rate vector from the gyros (bias-corrected)	rad/s
$\mathbf{a}$	rotation vector $\boldsymbol{\omega}\Delta t$; $\alpha = \|\mathbf{a}\|$ is the turn angle	rad
$[\mathbf{a}]_\times$	skew-symmetric matrix $\begin{pmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{pmatrix}$; $[\mathbf{a}]_\times\mathbf{b} = \mathbf{a} \times \mathbf{b}$	rad

WORKED EXAMPLE — one gyro step

Given: yaw rate $\omega_z = 0.0842$ rad/s (from a real IMU message in the seminar), $\Delta t = 0.005$ s (200 Hz). Start level, heading $\psi = 0$.

Step 1

$\alpha = 0.0842 \times 0.005 = 0.000421$ rad $= 0.0241°$, all about z

Step 2

For rotation purely about z, Rodrigues reduces to $\mathbf{R}_z(\alpha)$: $\cos\alpha = 0.99999991$, $\sin\alpha = 0.000421$

Step 3

New heading $\psi = 0.0241°$. If the rate stays constant, 200 such steps (1 s) add up to $200 \times 0.000421 = 0.0842$ rad $= 4.82°$

Result: the strapdown computer repeats this tiny multiplication 200 times per second to track attitude.

6.7 Why tiny angle errors matter

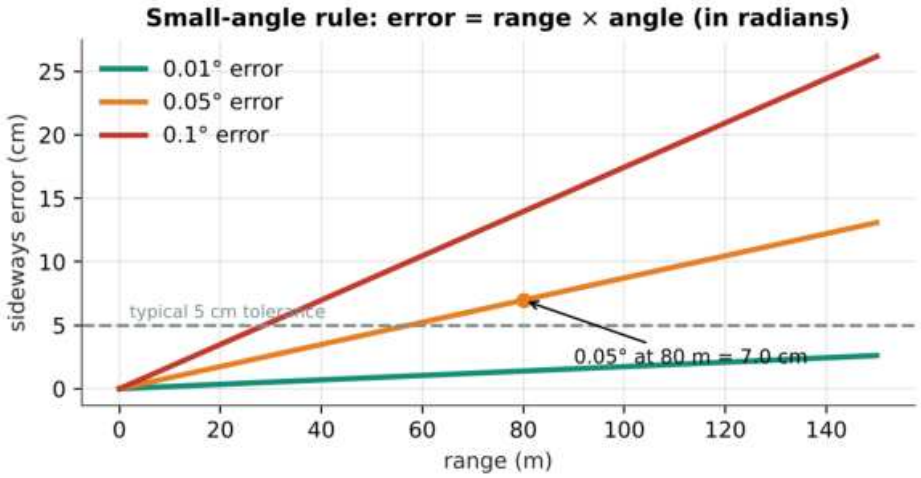


Figure 6.5: Lateral error versus range for three residual angle errors.

WORKED EXAMPLE — 0.05° heading error at 80 m

Step 1 — radians

$0.05 \times \pi/180 = 0.0008727$ rad

Step 2 — small-angle rule

$80 \times 0.0008727 = 0.0698$ m

Result: 7.0 cm. At 100 m: 8.7 cm. A 0.1° error at 100 m: 17.5 cm — bigger than most engineering tolerances.

TRY IT YOURSELF

6.1 Rotate $(10, 0)$ by 90°; then rotate $(0, 10)$ by $-90°$. What do you notice?
6.2 A vehicle drives with azimuth 135° (southeast). What is its ENU yaw?
6.3 What heading error produces 2 cm lateral error at 40 m?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," of the Reference Book, starting on page 163.

SOURCES FOR THIS MODULE

ROS REP-103 (FLU body frame, ENU world frame); Groves, P. D. (2013), *Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems*, 2nd ed. (rotations, FRD/NED conventions); Shan *et al.* (2020), eq. (6), (12).